

## Section 1 - Identification

**Product Name** Aluminum oxide, NanoArc™ AL-0450, 50% in water, colloidal dispersion with dispersant

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>44925</b>                                                                                         |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as not hazardous according to criteria of Safe Work Australia.

**Physical hazards**  
No hazards identified

**Health hazards**  
No hazards identified

**Environmental hazards**  
No hazards identified

**Label Elements** None required

**Other information**

## Section 3 - Composition and Information on Ingredients

| Component              | CAS No    | Weight % |
|------------------------|-----------|----------|
| Aluminum oxide         | 1344-28-1 | 50       |
| Water                  | 7732-18-5 | 40-50    |
| Proprietary dispersant | N/A       | 1-15     |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention immediately if symptoms occur.                                                                                                                                                                                                               |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                                                   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.                                                                                                                                                                 |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                         |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                       |
| <b>Self-Protection of the First Aider</b>  | No special precautions required.                                                                                                                                                                                                                                                        |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                    |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                  |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

Water.

### Hazardous Decomposition Products

Fumes of aluminum or aluminum oxide.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

#### Environmental Precautions

Should not be released into the environment.

#### Methods for Containment and Clean Up

##### Clean-up methods - small spillage

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

##### Clean-up methods - large spillage

Not applicable, packaged goods.

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

#### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

#### Conditions for Safe Storage, Including any Incompatibilities

Corrosives area. Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

#### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component      | Australia                 | New Zealand WEL           | ACGIH TLV                | The United Kingdom                                                                                                                        | Germany                                                                                                                                                                                                                        |
|----------------|---------------------------|---------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aluminum oxide | TWA: 10 mg/m <sup>3</sup> | TWA: 10 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | STEL: 30 mg/m <sup>3</sup> 15 min<br>STEL: 12 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 4 mg/m <sup>3</sup> 8 hr | TWA: 1.25 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 10 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 4 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 1.5 mg/m <sup>3</sup> (8 Stunden). MAK |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Exposure Controls

##### Engineering Measures

None under normal use conditions.

## Personal protective equipment

### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Natural rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |                 |                       |
| Neoprene       |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Skin and body protection

Long sleeved clothing

### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

### Recommended Filter type: Recommended half mask:-

Particle filter (or AUS/NZ equivalent)  
Particle filtering: EN149:2001 (or AUS/NZ equivalent)

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## Environmental exposure controls

No information available.

# Section 9 - Physical and Chemical Properties

## Information on basic physical and chemical properties

### Appearance

#### Physical State

Liquid dispersion

#### Odor

No information available

#### Odor Threshold

No data available

#### pH

No information available

#### Melting Point/Range

No data available

#### Softening Point

No data available

#### Boiling Point/Range

No information available

#### Flash Point

No information available

**Method -** No information available

#### Evaporation Rate

No data available

#### Flammability (solid,gas)

Not applicable

Liquid

#### Explosion Limits

No data available

#### Vapor Pressure

23 hPa @ 20 °C

#### Vapor Density

No data available

(Air = 1.0)

#### Specific Gravity / Density

No data available 1-1.4

#### Bulk Density

Not applicable

Liquid

#### Water Solubility

Immiscible

#### Solubility in other solvents

No information available

#### Partition Coefficient (n-octanol/water)

#### Autoignition Temperature

No data available

#### Decomposition Temperature

No data available

#### Viscosity

No data available

**Explosive Properties** No information available  
**Oxidizing Properties** No information available

**Other information**

**Molecular Formula** Al<sub>2</sub> O<sub>3</sub>  
**Molecular Weight** 101.96

## Section 10 - Stability and Reactivity

**Reactivity** Yes  
**Stability** Stable under normal conditions.  
**Conditions to Avoid** Exposure to moist air or water, Exposure to moisture.  
**Incompatible Materials** Water.  
**Hazardous Decomposition Products** Fumes of aluminum or aluminum oxide.  
**Hazardous Polymerization** No information available.

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

**(a) acute toxicity;**

**Oral** Based on available data, the classification criteria are not met  
**Dermal** No data available  
**Inhalation** No data available

#### Toxicology data for the components

| Component      | LD50 Oral                                    | LD50 Dermal | LC50 Inhalation                        |
|----------------|----------------------------------------------|-------------|----------------------------------------|
| Aluminum oxide | > 5000 mg/kg ( Rat )<br>(OECD Guideline 401) |             | > 2.3 mg/l 4 h<br>(OECD Guideline 403) |
| Water          | LD50 > 90 mL/kg ( Rat )                      |             |                                        |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

**Respiratory** No data available  
**Skin** No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component | Australia | New Zealand | New South Wales | Western Australia | IARC | EU | UK | Germany |
|-----------|-----------|-------------|-----------------|-------------------|------|----|----|---------|
|-----------|-----------|-------------|-----------------|-------------------|------|----|----|---------|

|                |  |  |  |  |  |  |  |                     |
|----------------|--|--|--|--|--|--|--|---------------------|
| Aluminum oxide |  |  |  |  |  |  |  | Cat. 2 (Fibre dust) |
|----------------|--|--|--|--|--|--|--|---------------------|

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## Section 12 - Ecological Information

|                                        |                                                                                                           |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Ecotoxicity effects</b>             | .                                                                                                         |
| <b>Persistence and Degradability</b>   |                                                                                                           |
| <b>Persistence</b>                     | Immiscible with water.                                                                                    |
| <b>Bioaccumulative Potential</b>       | May have some potential to bioaccumulate                                                                  |
| <b>Mobility</b>                        | Spillage unlikely to penetrate soil. Is not likely mobile in the environment due its low water solubility |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                 |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance                                            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance                                            |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                  |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                              |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |

## Section 14 - Transport Information

IMDG/IMO Not regulated

ADG Not regulated

**IATA** Not regulated

**Environmental hazards** No hazards identified

**Special Precautions** No special precautions required

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                  | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|----------------------------|-------------------------------------------------------------|------------------------|
| Aluminum oxide - 1344-28-1 | Present                                                     | -                      |
| Water - 7732-18-5          | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### **International Inventories**

| Component      | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCs | ISHL | IECSC | KECL     |
|----------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Aluminum oxide | X    | X     | 215-691-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-01012 |
| Water          | X    | X     | 231-791-2 | -      | X    | X   | -    | X     | X    |      | X     | KE-35400 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**  
Not applicable.

| Component              | CAS No    | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------|-----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Aluminum oxide         | 1344-28-1 | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Water                  | 7732-18-5 | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Proprietary dispersant | N/A       | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH** Not applicable

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**  
**Physical hazards** On basis of test data



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|                       |                    |
|-----------------------|--------------------|
| Health Hazards        | Calculation method |
| Environmental hazards | Calculation method |

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

|                  |                 |
|------------------|-----------------|
| Revision Date    | 19-Nov-2022     |
| Revision Summary | Not applicable. |

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet